

Dev Container Side-by-Side Runbook

Professional reference for running different Java versions in parallel with isolated IntelliSense and predictable startup behavior in VS Code.

Two VS Code windows

Container-isolated Java language server

Validated with runtime + IntelliSense checks

Demo examples: Java 17 and Java 21

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Outcome Summary

- Both Java contexts launch from a single script.
- Each window attaches to its own container image.
- Java version and API intelligence are correctly isolated.
- Pattern is reusable for any supported version pair.

Java 17

Separate runtime + IntelliSense behavior

Java 21

Separate runtime + IntelliSense behavior

2 Windows

Side by side, no SDK conflict

Launch Script

```
./launch-demo.ps1
```

Context Paths

- .devcontainer/java17
- .devcontainer/java21

Major Benefit: Host JDK installation is not required. Compile, run, and IntelliSense use the JDK inside each container.

Required Configuration

Dev Container wrappers

- .devcontainer/java17/.devcontainer/devcontainer.json
- .devcontainer/java21/.devcontainer/devcontainer.json

Each wrapper defines image, workspace mount, and workspace folder for stable resolution.

Host settings alignment

Setting	Expected Value
dev.containers.dockerPath	Rancher Desktop docker.exe path
remote.containers.dockerPath	Rancher Desktop docker.exe path
dev.containers.executelnWSL	false
remote.containers.executelnWSL	false

Validation Procedure

This procedure is version-agnostic. Java 17 and Java 21 are used below as demo examples.

Step 1: Runtime check

Run in each container terminal:

```
java -version
```

- Example A (Java 17 demo) should report version 17.
- Example B (Java 21 demo) should report version 21.

Step 2: IntelliSense split check

Use this exact line in both windows:

```
List.of(1,2,3).reversed();
```

- Example A (Java 17 demo): `reversed()` should be undefined (error).
- Example B (Java 21 demo): `reversed()` should be supported (no error).

This pair of checks confirms both runtime and language tooling are bound to the correct container context.

Screenshot Evidence

Evidence shown below is from the Java 17/21 demo pair to illustrate the general approach.

The screenshot shows the VS Code interface with a Dev Container for Java 17. The Explorer on the left shows the project structure. The Main.java file is open in the editor, showing a simple Java class. The Terminal at the bottom shows the output of the `java -version` command, which reports OpenJDK 17.0.16. The status bar at the bottom indicates the container is using Java 17.

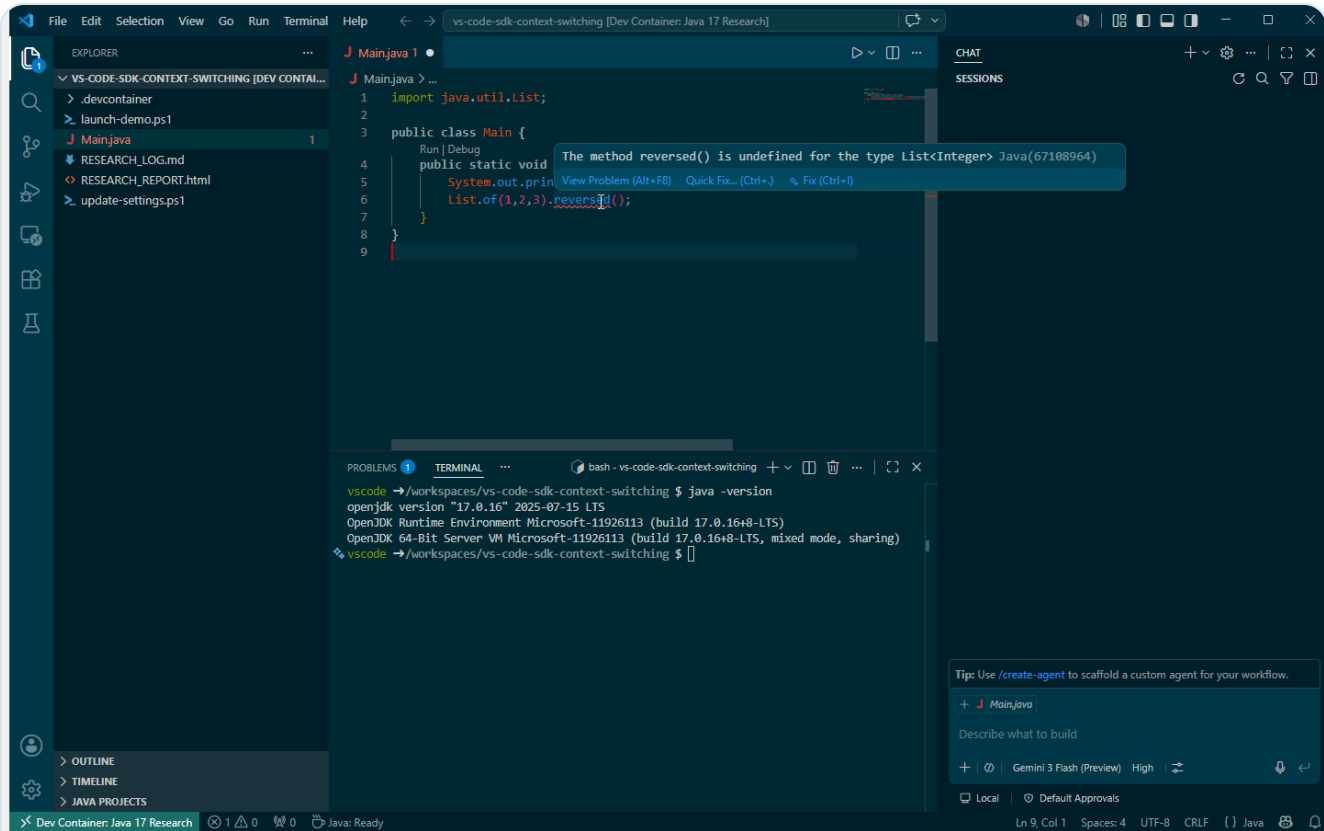
```
File Edit Selection View Go Run Terminal Help  
vs-code-sdk-context-switching [Dev Container: Java 17 Research]  
EXPLORER  
VS-CODE-SDK-CONTEXT-SWITCHING [DEV CONTAINER...]  
  .devcontainer  
  launch-demo.ps1  
  Main.java  
  RESEARCH_LOG.md  
  RESEARCH_REPORT.html  
  update-settings.ps1  
Main.java  
1 public class Main {  
2     public static void main(String[] args) {  
3         System.out.println("Java Version: " + System.getProperty("java.version"));  
4     }  
5 }  
6  
PROBLEMS TERMINAL  
bash - vs-code-sdk-context-switching  
vscode -> /workspaces/vs-code-sdk-context-switching $ java -version  
openjdk version "17.0.16" 2025-07-15 LTS  
OpenJDK Runtime Environment Microsoft-11926113 (build 17.0.16+8-LTS)  
OpenJDK 64-Bit Server VM Microsoft-11926113 (build 17.0.16+8-LTS, mixed mode, sharing)  
vscode -> /workspaces/vs-code-sdk-context-switching $  
Dev Container: Java 17 Research 0 0 0 Java: Ready  
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF () Java
```

Java 17 runtime verification

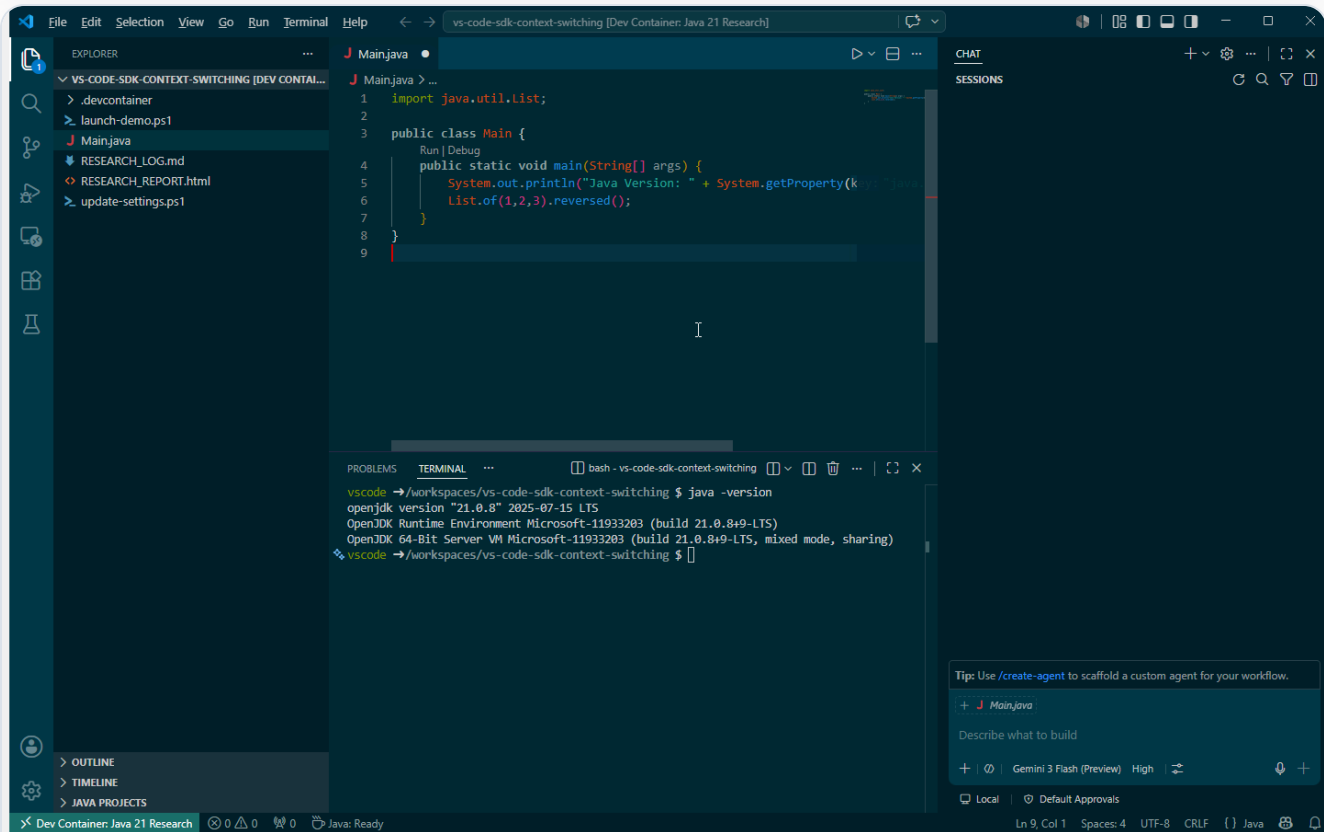
The screenshot shows the VS Code interface with a Dev Container for Java 21. The Explorer on the left shows the project structure. The Main.java file is open in the editor, showing a simple Java class. The Terminal at the bottom shows the output of the `java -version` command, which reports OpenJDK 21.0.8. The status bar at the bottom indicates the container is using Java 21.

```
File Edit Selection View Go Run Terminal Help  
vs-code-sdk-context-switching [Dev Container: Java 21 Research]  
EXPLORER  
VS-CODE-SDK-CONTEXT-SWITCHING [DEV CONTAINER...]  
  .devcontainer  
  launch-demo.ps1  
  Main.java  
  RESEARCH_LOG.md  
  RESEARCH_REPORT.html  
  update-settings.ps1  
Main.java  
1 public class Main {  
2     public static void main(String[] args) {  
3         System.out.println("Java Version: " + System.getProperty("java.version"));  
4     }  
5 }  
6  
PROBLEMS TERMINAL  
bash - vs-code-sdk-context-switching  
vscode -> /workspaces/vs-code-sdk-context-switching $ java -version  
openjdk version "21.0.8" 2025-07-15 LTS  
OpenJDK Runtime Environment Microsoft-11933203 (build 21.0.8+9-LTS)  
OpenJDK 64-Bit Server VM Microsoft-11933203 (build 21.0.8+9-LTS, mixed mode, sharing)  
vscode -> /workspaces/vs-code-sdk-context-switching $  
Dev Container: Java 21 Research 0 0 0 Java: Ready  
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF () Java
```

Java 21 runtime verification



Java 17 IntelliSense: reversed() undefined



Java 21 IntelliSense: reversed() supported

Known Risk and Recovery

Rancher Desktop backend instability can break launches even when repository configuration is correct.

1. Confirm Rancher Desktop is fully started.
2. Validate Docker health with `docker.exe` version.
3. Rerun `launch-demo.ps1`.

Host Machine Requirements

- No host JDK install is needed for this approach.
- Required: Docker runtime (for example Rancher Desktop).
- Required: VS Code with Dev Containers extension.

JDK lifecycle management is handled in container images/configuration, not on the host.

Documentation generated from validated project state and captured session evidence.